INCH-POUND
MIL-I-24718/2
6 April 1990

MILITARY SPECIFICATION SHEET

INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING
REPLENISHMENT RESIN, EPOXY-LEWIS ACID, SLIGHTLY THIXOTROPIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24718.

Resin: The product shall consist of primarily epoxy resin with a Lewis acid type catalyst and appropriate filler and thixotropes. The product shall be compatible with resin samples provided by the acquisition activity and meet all of the requirements specified herein. The resin samples provided shall be of the same generic class (for example, epoxy-Lewis acid).

Thermal class: The cured resin mixtures (see compatibility section below) shall have thermal endurance that is not less than the thermal endurance of the sampled resin when tested in accordance with MIL-I-24718.

Rheology: The rheological properties of the resin mixtures shall be as specified and suitable for VPI of form and random wound electrical coils and windings.

Compatibility: The mixing tolerance of the catalyzed product shall be determined by using new resin prepared in accordance with MIL-I-24718 and resin supplied by the acquisition activity. The new resin shall be added to the supplied resin at 23°C, stirring slowly until mixtures of 9:1, 3:1, 1:1, 1:3, and 1:9 by weight are obtained. The mixtures shall then be examined for curdling, precipitation, or separation and shall be examined again after 24 hours. The resulting mixtures shall be homogeneous. Each mix shall meet the first article inspection as specified herein. The replenishment product shall also be tested without addition of any other material.

Restrictions: The epoxy-Lewis acid insulating resin formulations are not suitable for use on high voltage AC equipment (>5KV).

AMSC N/A FSC 5970

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Requirements. The following requirements shall be met for 100 percent replenishment product and all mixtures of replenishment product with supplied resin.

First article inspection. First article inspection shall be as specified in MIL-I-24718 and table I herein.

TABLE I. First article inspection.

Characteristics	Unit	Requirements
Gel time	Minutes	Within manufacturer's specified range
Catalyzed storage life	Months at 20°C, minimum	9
Vapor pressure	mm Hg, maximum	0.5
Weight loss on cure	Average percent, maximum	3.0
Flash point	°C, minimum	93
Viscosity		
at 2 r/min	Average centipoise, maximum	18,000
at 20 r/min	Average centipoise, maximum	8,000
Shear thinning index	Minimum	1.10
Build	Mils, minimum	1.5
	Mils, maximum	3.0
Resin retention	Weight percent, minimum	75.0
Dielectric strength		
at 96/23/50	Volts per mil, minimum average per panel	1,000
at 96/23/96	Percent decrease from 96/23/50 value, maximum	25
at 24/23/water	Percent decrease from 96/23/50 value, maximum	25
Dissipation factor		
at 23°C	Percent, maximum	3.0
at 150°C	Percent, maximum	NA
Dielectric constant		
at 23°C	Maximum	5.0
at 150°C	Maximum	NA
Bond strength		
at 23°C	Pounds force, minimum	30.0
at 150°C	Pounds force, minimum	5.0
Thermal endurance, all classes	Thermogravimetric analysis comparison with sampled resin, °C difference at any weight loss	±5
Hydrolytic stability	Percent decrease in bond strength, maximum	50
Hardness	Type D, minimum	72
Chemical resistance		
Hydraulic fluid	Percent decrease in bond strength, maximum	25
Lubricating oil	Percent decrease in bond strength, maximum	25

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TABLE I. First article inspection - Continued.

Characteristics	Unit	Requirements
Cleaning fluid	Percent decrease in bond strength, maximum	25
Distilled water	Percent decrease in bond strength, maximum	25
Specific gravity	Unitless	Within manufacturer's specified range

Quality conformance inspection. Quality conformance inspection shall be as specified in MIL-I-24718 and table II herein:

TABLE II. Quality conformance requirements. 1/

Characteristics	Unit	Requirements
Detergent solution	Percent decrease in bond strength, maximum	25
Weight loss on cure	Average percent, maximum	3.0
Gel time	Percentage of qualification value, plus or minus	10
Viscosity at 2 r/min	Average centipoise, maximum	20,000
Build	Percentage of qualification value, plus or minus	15
Resin retention	Weight percent, minimum	75.0
Specific gravity	Percentage of qualification value, plus or minus	10.0
Dielectric strength at 24/23/50	Volts per mil, minimum	1,000
Hardness	Percentage of qualification value, plus or minus	10.0
Bond strength at 25°C	Pounds force, minimum	30.0

1/ Periodic conformance requirements shall be measured on the replenishment product only; mixing with sampled resin is not necessary.

Intended use: The product shall be used as a replenishment resin for MIL-I-24718/1 type of VPI resins only after approval from NAVSEA. It shall be tested alone and mixed with sample storage resin supplied by the VPI facility.

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Part or Identifying Number (PIN): The PIN shall be M24718/2-(dash number indicating thermal class). For example, M24718/2-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
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Preparing activity:
Navy - SH
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-24718/2	2. DOCUMENT DATE (YYMMDD)

DOCUMENT TITLE

INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING: GENERAL SPECIFICATION FOR

3. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

4. REASON FOR RECOMMENDATION

5. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

6. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. C.Y. Lu (SEA 56Z24) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 703-602-3123	(2) AUTOVON 8-332-3123
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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